

LINCOLN, IL, USA

WMO: 744672

Lat: 40.158N

Lon: 89.335W

Elev: 182

StdP: 99.16

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 04862

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.1	-14.1	-21.3	0.6	-15.6	-18.9	0.7	-13.3	10.4	2.4	9.1	2.7	2.0	290	0.456

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	34.1	24.5	32.7	24.0	31.3	23.4	26.7	31.1	25.7	30.3	24.8	29.5	2.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.5	21.2	29.3	24.3	19.7	28.5	23.3	18.4	27.4	84.6	30.3	80.0	30.3	76.1	29.9	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.8	6.9	DB	-21.4	36.1	3.5	2.1	-23.9	37.6	-26.0	38.9	-28.0	40.1	-30.6	41.6	(4)
(5)			WB	-21.7	28.2	3.4	1.4	-24.1	29.2	-26.0	30.0	-27.9	30.7	-30.4	31.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	-2.7	-1.2	5.8	12.2	18.2	23.0	24.3	23.2	20.0	13.0	5.8	0.2	(6)	
	DBStd	10.81	6.64	6.26	6.10	4.99	4.89	3.31	3.11	2.94	4.07	5.08	5.37	5.63	(7)	
	HDD10.0	1383	394	316	161	33	3	0	0	0	0	26	146	306	(8)	
	HDD18.3	2999	651	547	391	192	66	3	1	2	28	181	376	562	(9)	
	CDD10.0	2069	2	3	30	97	256	389	444	408	300	118	21	2	(10)	
	CDD18.3	642	0	0	2	7	61	142	186	152	78	14	1	0	(11)	
	CDH23.3	6390	0	0	18	97	601	1397	1914	1414	804	142	3	0	(12)	
(13) Wind	CDH26.7	2355	0	0	2	14	186	527	778	518	300	31	0	0	(13)	
	WSAvg	2.7	3.2	3.2	3.5	3.5	2.9	2.2	1.6	1.6	1.8	2.4	3.1	3.0	(14)	
	PrecAvg	983	47	37	76	97	109	100	110	99	82	72	71	63	(15)	
	PrecMax	1264	157	87	187	251	323	239	405	218	212	218	192	242	(16)	
	PrecMin	738	1	3	28	15	27	17	16	14	12	6	8	9	(17)	
	PrecStd	144	37	22	39	53	59	53	71	51	49	47	40	44	(18)	
	DB	16.2	18.6	25.9	28.3	32.8	35.1	37.2	35.3	34.9	29.9	22.5	17.5		(19)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	13.9	14.0	17.9	17.3	21.8	23.5	25.2	24.7	23.2	21.2	15.5	15.0		(20)	
	DB	12.6	14.7	21.7	26.3	30.6	33.0	34.0	32.8	32.5	27.1	19.1	13.4		(21)	
	MCWB	10.8	11.5	15.8	16.6	20.8	23.6	25.4	24.3	22.8	19.0	14.2	11.3		(22)	
	DB	9.2	11.0	18.2	23.7	28.8	31.6	32.5	31.3	30.1	24.8	16.9	10.9		(23)	
	MCWB	7.3	8.4	13.8	15.9	20.0	23.0	25.0	23.8	21.6	17.8	13.3	9.4		(24)	
	DB	6.1	7.4	15.7	21.6	27.1	29.9	30.9	29.7	27.8	22.1	14.3	8.1		(25)	
	MCWB	4.3	5.1	11.8	14.5	18.7	22.1	24.5	22.9	19.6	16.8	11.1	6.5		(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	14.7	15.6	18.7	19.8	24.0	26.6	28.7	27.8	25.5	22.9	17.7	15.3		(27)	
	MCDB	15.9	17.4	24.2	25.7	29.9	32.0	33.4	32.2	31.4	27.3	19.9	16.6		(28)	
	WB	11.2	11.9	16.7	18.6	22.6	25.3	27.2	26.4	24.1	21.4	15.6	12.1		(29)	
	MCDB	12.1	14.0	20.6	23.4	27.9	31.0	31.5	30.4	29.2	25.0	18.2	13.0		(30)	
	WB	7.5	8.5	14.9	17.3	21.5	24.3	25.9	25.4	22.9	19.8	13.7	9.1		(31)	
	MCDB	8.9	10.7	18.0	21.7	26.6	29.4	30.4	29.2	27.7	22.5	16.4	10.5		(32)	
	WB	4.3	5.3	12.4	15.7	20.4	23.4	25.1	24.2	21.9	17.6	11.6	6.6		(33)	
(34) Mean Daily Temperature Range	MCDB	6.1	7.3	14.9	20.2	25.2	28.2	29.9	28.2	26.2	21.0	14.1	7.9		(34)	
	MDBR	8.4	9.1	10.2	12.0	11.4	11.2	11.0	11.3	12.9	11.7	9.8	7.8		(35)	
	5% DB	11.5	13.3	13.9	15.6	13.4	12.9	12.2	13.1	14.9	15.0	13.3	11.2		(36)	
	MCWBR	9.5	10.2	8.9	8.2	5.8	5.6	5.5	5.9	6.1	7.7	9.0	9.4		(37)	
	5% WB	11.2	12.8	12.3	12.7	11.4	11.1	10.9	10.6	12.2	11.5	12.0	10.3		(38)	
	MCWBR	9.9	10.3	8.5	8.4	5.9	5.4	5.8	5.9	5.6	7.1	9.4	9.3		(39)	
	taub	0.300	0.312	0.349	0.388	0.433	0.441	0.458	0.445	0.407	0.358	0.310	0.295		(40)	
(41) Clear-Sky Solar Irradiance	taud	2.432	2.393	2.376	2.271	2.201	2.224	2.162	2.187	2.271	2.409	2.496	2.462		(41)	
	Ebn at Noon	865	905	904	885	849	840	821	821	830	840	849	843		(42)	
	Edh at Noon	80	96	110	130	143	140	148	140	121	94	75	72		(43)	
	RadAvg	1.84	2.72	3.68	4.80	5.49	6.27	6.25	5.51	4.67	3.19	2.13	1.55		(44)	
(45) All-Sky Solar Radiation	RadStd	0.21	0.25	0.27	0.35	0.44	0.46	0.36	0.35	0.35	0.36	0.19	0.20		(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+102	+71

Nomenclature: See separate page